MILITARY SPECIFICATION

EXTINGUISHER, FIRE, DRY CHEMICAL AND AQUEOUS FILM FORMING FOAM: SELF-CONTAINED

This specification is approved for use by the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirement for one type twin-agent fire extinguisher.

2. APPLICABLE DOCUMENTS

2.1 *Government documents*

2.1.1 *Specifications, standards, and handbooks*. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-D-1407 - Dry Chemical, Fire Extinguishing, Potassium Bicarbonate

BB-N-411 - Nitrogen, Technical

PPP-B-601 - Boxes, Wood, Cleated Plywood

PPP-B-621 - Box, Wood, Nailed and Lock Corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps (LMA-1), Headquarters U.S. Marine Corps, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

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MIL-P-116 - Preservation, Methods Of

MIL-T-704 - Treatment and Painting of Materiel

MIL-C-5015 - Connector, Electrical, Circular Threaded, An Type, General Specification For

MIL-F-24385 - Fire Extinguishing Agent, Aqueous Film Forming Foam (AFFF), Liquid Concentrate, For Fresh and Sea Water

MIL-E-52798 - Enamel, Alkyd, Camouflage

MIL-F-82129 - Fuel System, Amphibious Assault and Tactical Airfield Dispensing; General Specification For

STANDARDS

MILITARY

MIL-STD-129 - Marking For Shipment And Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-794 - Part And Equipment, Procedures for Packaging And Packing Of

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION SPECIFICATIONS

Specification DOT-3AA, Contained in Title 49 Transportation, Code of Federal Regulations Part 100 to 199.

Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS
Code for Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION

Pamplet No. 412, Evaluating Foam Fire Equipment for Aircraft Rescue and Fire Fighting Vehicles (1974 Edition)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch St., Boston, Mass. 02110.)

Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specifiction shall take precedence.

3. REQUIREMENTS

3.1 First article. Prior to starting production, the supplier shall furnish a first article for examination and test to prove that his production methods and choice of design detail will produce fire estinguishers which comply with the requirements of this specification. Examination and tests shall be as specified herein and shall be subject to surveillance and approval by the Government.

3.2 Parts and materials. Parts and materials shall be as specified herein. Parts and materials not specified shall be of the best quality and entirely suitable for the purpose intended, with adequate strength for safety and reliability and shall be corrosion resistant.

3.2.1 Nonstandard parts and materials. When it has been determined that requirements cannot be met by parts and materials specified herein, immediate action shall be taken by the supplier to inform the contracting officer by submission of detailed data, requesting approval of nonstandard parts and materials.

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3.2.2 Metals. All metal parts shall be of corrosion resistant material or treated in a manner to render them adequately resistant to corrosion. Dissimilar metals shall not be used in intimate contact with each other, unless they have been suitably protected against electrolytic corrosion.

3.2.3 Nonmetallic materials. Nonmetallic materials which will be adversely affected by continued use with fire extinguishing agents specified herein shall not be used.

3.3 Design and construction. This fire extinguisher shall be a twin-agent type using fire extinguishing agents and charge as follows:

(a) Aqueous film forming foam (AFFF) conforming to MIL-F-24385, type 6.

(b) Potassium bicarbonate dry chemical conforming to O-D-1407.

(c) Nitrogen charge conforming to type I, class I, grade B, of BB-N-411.

The fire extinguisher shall be skid-mounted and shall be designed and constructed to permit easy inspection, operation, and recharging and shall consist of an AFFF agent tank, a dry chemical agent tank, nitrogen cylinders, piping, pressure regulators, discharge hoses and nozzles, hose reel and other equipment specified herein. The fire extinguisher shall be designed primarily to utilize the potassium bicarbonate dry chemical specified in (b) above, but shall also be capable of utilizing the nitrogen charge in (c) above without changing any of the mechanical components.

3.3.1 Twin-agent operating principle. The AFFF solution and the dry chemical agent shall be stored unpressurized in individual tanks. The expellant energy required to apply the agents shall be provided by nitrogen under pressure. When in use, nitrogen flow from the cylinders shall enter the agent tanks, pressurizing the active AFFF solution and simultaneously fluidizing and pressurizing the dry chemical agent. The agents distribution system shall begin at the outlets of both tanks and shall consist of a twin passage hose terminating in twin hand-operated valves and nozzles for controlling the agents' flow and direction. As the valves are opened, either individually or simultaneously, the agents shall flow from the pressurized tanks through the hose and from the nozzles.

3.3.2 AFFF agent tank. The AFFF agent tank shall have a capacity of not less than 100 gallons. It shall be designed and constructed in accordance with the ASME Code for Pressure Vessels, Part UG of Division I of Section VIII and shall be National Board stamped as suitable for use of 250 pounds-force per square inch (psi) working pressure in temperature ranges of -20°F to +130°F. It shall be constructed of either stainless

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steel, or a carbon steel lined with coal tar epoxy compatible with AFFF and salt or fresh water solutions. A reseating pressure-relief valve meeting the requirements of UG-125 through UG-134 of the ASME Pressure Vessel Code, set to operate at 250 psi (110 percent of operating pressure) shall be provided. It shall be capable of venting from the void volume of the tank at the maximum possible gas flow into the tank. A readily accessible 4-inch diameter fill opening shall be provided in compliance with UG-36 through UG-46 of the ASME Pressure Vessel Code. It shall be located in a position which allows the liquid level to be measured with a direct reading dipstick. The agent outlet opening in the tank shall be no smaller in size than the internal diameter of the discharge hose. Valving shall be provided to route the nitrogen supply directly to the hose line inlet, bypassing the AFFF agent tank, in order to purge the hose line of residual solution. A vent line with a "quarter-turn" valve shall be provided on the top of the tank to permit "blowdown" to insure zero pressure in the tank prior to opening the long handled fill cap. A temperature relief valve set for 212°F shall be installed in the tank, positioned so that it will be in contact with the AFFF solution until the tank is empty. Provision shall be made for draining the tank for storage.

3.3.2.1 AFFF tank heater system. A "blanket" type heater assembly shall be installed on the AFFF tank assembly, including a suitable type of insulating foam covering. This heater shall be thermostatically controlled to maintain temperature of the AFFF solution above a minimum temperature of 40°F in an ambient temperature of -40°F. An overheat safety device shall automatically shut off the heater when the AFFF solution reaches 80°F ±5°F. The heater system shall be compatible with a 3-phase 60-hertz (Hz), 4-wire (wye connected) 120-208 volts alternating current (Vac) power supply with a military standard connector conforming to MIL-C-5015. A 1-inch pipe fitting (plugged) shall be included in the AFFF tank to accept a thermostat or alternate type of heating element.

3.3.2.2 Dipstick and funnel. A direct-reading dipstick and a funnel suitable for filling the AFFF tank shall be provided.

3.3.3 Dry chemical agent tank. The dry chemical agent tank shall be capable of containing at least 450 pounds of "freshfill" (untamped) potassium bicarbonate dry chemical agent. The tank shall be designed and fabricated in accordance with Part UG of Division I of Section VIII of the ASME Code for Pressure Vessels for working pressure of 250 psi and shall be National Board stamped. It shall be suitable for use in the temperature range of -20°F to +130°F. A reseating pressure relief valve satisfying Parts UG-125 through UG-134 of the ASME Pressure Vessel Code, capable

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of relieving 250 psi (110 percent of design pressure), shall be provided. This relief valve shall be mounted to vent the void volume above the agent level at the maximum gas flow into the tank. A readily accessible fill opening at least 4 inches in diameter shall be provided in compliance with UG-46 of the ASME Pressure Vessel Code. A nitrogen distribution system shall be provided to fluidize, pressurize, and discharge the dry chemical agent at a flow of at least 5 pounds per second. Valving shall be provided to route the nitrogen supply directly to the hose inlet, by-passing the dry chemical tank, in order to purge the hose of residual dry chemical and prevent hose packing. A vent line shall be provided in the top of the tank for "blowdown" purposes to insure zero pressure in the tank before opening the fill cap. This vent line shall exhaust into the dry chemical hose inlet.

3.3.4 Fill caps. The fill caps of both the AFFF dry chemical agent tanks shall be provided with rubber gaskets and safety vent holes so that each cap is pressure-vented while at least three and one-half threads are still engaged. The fill cap arrangement shall be such that the female threads and handles are part of the cap.

3.3.5 Nitrogen cylinders. A nitrogen gas cylinder(s) shall be provided for each of the two agent tanks. The cylinders shall be of sufficient capacity to completely discharge the agents (at the rated delivery rates) and purge all lines of residual agents. These cylinders shall be mounted on racks. The cylinders shall be of shatter-proof construction in accordance with the DOT Regulation 3AA. Pressure gages (0-4000 lb_f/in^2) for indicating the cylinder gas pressure and quick-opening lever operated valves requiring only 90° rotation of the lever to release the nitrogen gas shall be provided. These quick operating valves shall be capable of being easily reset manually. Mounting racks for these cylinders shall provide for positioning of the cylinders in the horizontal position with the quick-opening valves at the operating end of the unit.

3.3.6 Nitrogen pressure regulators. Two pressure regulators shall be provided (one for each agent tank). Pressure regulators shall be designed to automatically reduce the high pressure gas cylinder pressure and maintain the expellant gas pressure at 230 to 250 psi, i.e., at or below the designed operating pressures of the agent tanks. The gas flow rate of the regulators and the gas distribution system of both tanks shall reach a pressure level of 80 percent of the operating pressure within 10 seconds after the cylinder valve is opened. The regulators shall be shall be sealed after final adjustment. In normal operation, gas from the nitrogen cylinders shall pass through the regulators, test valves, check valves, agent tanks, hose reel and handlines, and shall be controlled at the handline nozzles.

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3.3.7 Discharge hoses. The handline shall be constructed of two noncollapsible, 3-braid neoprene lined and jacketed hoses 150 feet long. Both hoses shall have not less than 1-inch inside diameter and shll be fastened together with a smooth, abrasion-resistant woven polyester jacket running the full length of the hoseline. Strands of the woven jacket shall be prevented from unraveling by bonding a 1-7/8-inch-wide neoprene band to the outer jacket. The handline shall have a nominal working pressure of up to 400 psi and a bursting pressure of 1500 psi. The material used must not be adversely affected by long exposure to the chemical agents.

3.3.8 Nozzles. The dry chemical and AFFF nozzles shall be close-coupled so that the nozzles may be easily operated and directed by one man. The nozzles shall be individually or simultaneously operated by an operator wearing fire-fighting protective clothing. Nozzles shall be mounted in a manner to prevent interference with either discharge stream. The nozzle assembly shall be of nonferrous construction, such as anodized aluminum, stainless steel, or chrome plated brass. Nozzles shall offer no obstruction to the flow in the passage ways, when the shut-off handles are rotated 90° from the full-off to the full-open position. The dry chemical nozzle tip shall give a semidispersed pattern at flow rates up to 6 pounds per second. The effective reach shall be 40 to 50 feet when the wind is less than 5 miles per hour (mph). The AFFF nozzle tip shall provide an effective reach of 45 to 50 feet, a 12-foot pattern width, and a foam expansion ratio of five to one when used with 6-percent AFFF solution. Range and pattern shall be determined with a minimum 100-psi inlet pressure, nozzle inclination of 15° above horizontal, and a nominal flow rate of 50 gallons per minute (gpm). The expansion ratio shall be measured as defined in NFPA Pamphlet No. 412. A suitable nozzle mount shall be provided on the extinguisher for holding nozzles when not in operation.

3.3.9 Hose reel. One manual rewind twin hose reel shall be provided for both dry chemical agent and AFFF agent discharge hoses. It shall provide for easy one man rapid pay-out and take-up storage of 150 feet of twin hose. Normal flow through the nozzle is required when any portion of the 150-foot hose is withdrawn from the reel.

3.3.9.1 Hose reel, frame assembly. The hose reel frame assembly shall be constructed of steel material of sufficient strength to withstand the cranking load and prevent deflection of the frame assembly and subsequent pinion and hose reel ring gear disengagement during hose unwinding and rewinding operations.

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3.3.10 Portable remote hose station. The portable remote hose station shall consist of a hose reel assembly, two 150-foot lengths of discharge hose, nozzle assembly, hose reel frame assembly, two 12-inch diameter wheels for mobility and a "push"-"pull" handle assembly. The hose reel assembly shall be identical in construction to the hose reel assembly described in 3.3.9. The discharge hoses shall be identical in construction and meet the same requirements described in 3.3.7. Additionally, the remote station discharge hoses shall be equipped with two quick-release fittings for coupling the hose to the nozzle ends of the two main discharge hoses. The remote hose reel frame assembly shall be identical to construction to the hose reel frame assembly described in 3.3.9.1. The frame assembly shall be mounted on a steel frame of sufficent strength to support the total combined weight of the hose reel assembly, discharge hoses, nozzles and frame assembly. For ease of mobility, the remote hose station shall be mounted on two steel rim, rubber tire wheels attached to a steel axle. A steel pipe handle assembly shall be constructed to either "push" or "pull" the hose station to a remote location. Figure 1 shall be used as guidance in the design and fabrication of the portable remote hose station.

3.3.11 Piping. Piping shall be designed to accommodate the flow rates specified herein without excessive friction losses. The piping system design shall inherently prevent loosening of joints and rupturing of pipes or connections under shock and vibration loads. No black or galvanized iron shall be used in contact with the AFFF agent. Potential galvanic corrosion points shall be eliminated by proper choice of materials.

3.3.12 Valves. Check valves shall be provided in the nitrogen supply system to prevent backflow of the extinguishing agents into the pressure regulators. Some means, such as valves or rupture discs, shall be used to prevent the agents from entering the hose lines until pressurization occurs. All valves used in the control of the dry chemical agents must be suitable for use with the agents in 3.3. These valves shall be designed to withstand, without failure, 125 continuous "ON/OFF" cycles during a nominal 5 pounds per second dry chemical agent flow. All valves shall have a size no less than the inside diameter of the connecting piping or hoses and shall be the "one-quarter" turn type. All valves shall be designed and installed to facilitate maintenance and replacement. All valves which will come in contact with the AFFF agent during normal operation shall be designed to show no signs of binding or seat deterioration after immersion in a 6-percent AFFF agent for 40 hours at a temperature of 100° ±10°F.

3.3.13 Castings and forgings. All castings shall be free from blowholes, porosity, hard-spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, mismatching and other defects which might affect the structural strength. Strength, and other essential physical and chemical properties of the castings and forgings, shall be adequate to meet the performance requirements specified herein.

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KEY	ITEM
1	BUSHING
2	HOSE ASSY
3	REEL ASSY
4	NIPPLE
5	HOSE ASSY
6	DRAWBAR ASSY
7	NUT
8	WASHER
9	ADAPTER
10	COUPLER
11	GASKET
12	U-BOLT
13	NUT
14	LOCKWASHER
15	SCREW
16	BOLT
17	FRAME ASSY
18	WASHER
19	WHEEL ASSY
20	BEARING
21	TIRE
22	WHEEL
23	FITTING
24	COTTER PIN
25	LOCKWASHER
26	CAP
27	PLUG

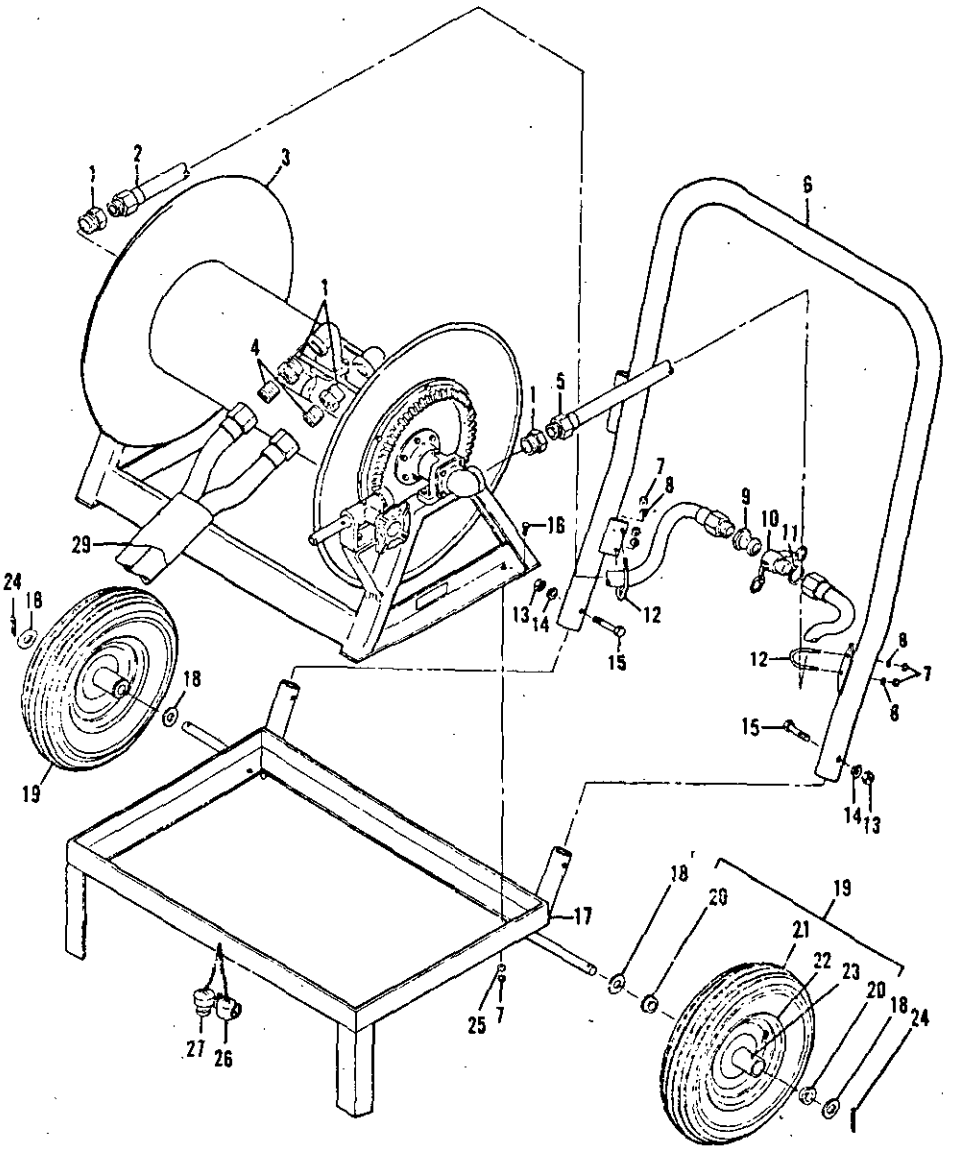


Figure 1. Portable Remote Hose Station

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3.3.14 Skid-mounting frame. The skid-mounting frame shall be constructed of electrically welded structural steel, shall be designed to withstand rough usage and shall permit forklift or crane (helicopter) lifting. The skid shall be provided with at least four lifting eyes to anchor the fire extinguisher to a truck-bed or aircraft floor and to facilitate lifting or towing. The shipping weight and center-of-gravity location shall be marked along each axis for both the charged and uncharged condition. The skids shall be provided with openings or insert forklift tines. Each opening shall be 3 inches by 12 inches. The distance between openings shall be 34 inches measured inside to inside. The openings shall be located so that the center of gravity of the fire extinguisher falls approximately midway between centers of the forklift openings.

3.3.15 Static electricity. A static wire of sufficient size to ground both nozzles to the unit and to prevent shocks caused by agent flow through the lines or nozzles shall be provided in the handline assembly.

3.3.16 Safety devices. The fire extinguisher shall be equipped with safety devices to adequately protect all hazards incident to their operation.

3.3.17 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.4. Maintainability. All subsystems, assemblies, and components shall be designed and assembled to afford maximum operational readiness with a minimum of preventive and corrective maintenance. Adjustments shall be readily accomplished by operating personnel without the need of special training or special tools. All components subject to wearout, deterioration, breakage, or loss of adjustment shall be readily accessible for replacement, repair, or adjustment. Nitrogen cylinder replacement and agent recharging operations shall be accomplished from the ground with a minimum of repiping or disassembling of components. Servicing from the bottom shall be avoided. There shall be no preventive maintenance requirement for servicing the dry chemical agent storage system to prevent compaction of the powder. The design of the dry chemical agent distribution system shall inherently provide the means for overcoming compacting problems without any manual operation.

3.5 Performance. The fire extinguisher shall be capable of the following performance characteristics.

3.5.1 Operational capability. The fire extinguisher shall be capable of achieving 80 percent of the operating pressure through the gas regulators and gas distribution system within 10 seconds after the nitrogen cylinder valves have been opened.

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3.6 Dimensions. The dimensions of the skid-mounted fire extinguisher shall not exceed 50 inches wide, 120 inches long and 60 inches high.

3.7 Weight. The weight of the fire extinguisher, fully charged, shall not exceed 4,700 pounds.

3.8 Treatment and paintings. All ferrous metal surfaces which do not possess a resistance to corrosion (including framing, dry chemical agent tank, AFFF agent tank, hose reel, and piping) shall be protected with an inorganic zinc coating. This method shall consist of a white metal sandblasting, a spray application of inorganic zinc coating with maximum of 3 to 5 mils dry film thickness followed by epoxy base primer with maximum of 1.2 to 1.8 mils dry film thickness. A color-coat finish of epoxy base with maximum of 1.5 to 3.0 mils dry film thickness shall be applied. This color-coat shall be forest green in accordance with MIL-E-52798. Any aluminum surfaces shall be treated in accordance with MIL-T-704. They shall not be sandblasted and shall be coated with zinc-chromate primer before applying the epoxy base color-coat. Finish thickness on the aluminum surfaces shall not exceed 3 to 5 mils dry film thickness. All stainless steel surfaces shall be etched (passitated) with a nitric acid solution, thoroughly rinsed with water, and prime coated with zinc-chromate primer prior to application of epoxy base color coating.

3.9 Marking. All plates, instruction and identification, shall be permanently mounted on the fire extinguisher at the rear so that they may be easily read by personnel intending to operate the fire extinguisher.

3.9.1 Identification plate. A metal nameplate provided by the procuring activity shall be permanently attached to the fire extinguisher. The identification marking shall be in accordance with MIL-STD-130, and shall contain the following information:

Nomenclature

Contract or Order No.

Manufacturer's Name or Trademark and Address

3.9.2 Identification bands. Two bands of reflectroized adhesive tape shall be attached to each agent tank. Each band shall be 2 inches wide with a minimum of 3 inches separation between bands. The AFFF agent tank bands shall be gold in color and shall bear the legent "AFFF" in black letters. The bands on the dry chemical tanks shall be purple in color and shall bear the legend "PKP" in white letters.

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3.9.3 Instruction plates. Instruction plates shall be provided showing the following:

(a) Complete and detailed activation instructions.

(b) Shutdown and maintenance instructions.

(c) Instructions concerning when and how to replace the nitrogen cylinders.

(d) Use of heaters and protection against subfreezing temperatures during storage.

(e) Refilling tables and instructions for all agents showing quantities required to refill tanks when compared with quantity of agent remaining in the tanks.

3.10 Workmanship. The extinguishers shall be uniform in quality and free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability or durability. Safety device shall be provided in accordance with manufacturing practices used in the production of this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)

(b) Acceptance (see 4.4)

(c) Packaging (see 4.5)

4.3 First article inspection. The first article shall be subjected to the examination specified in 4.6. Presence of one or more defects shall be cause for rejection. The first article shall be subjected to all tests in 4.7. Failure of any test shall be cause for rejection. Acceptance of the first article does not relieve the contractor of the requirement to comply with this specification.

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4.4 Acceptance inspection. Each fire extinguisher shall be subjected to examination in 4.6 and tests in 4.7.2, 4.7.3, and 4.7.5.2. Failure to pass any of these tests shall be cause for rejection.

4.5 Packaging. The preservation, packaging, packing and marking of each fire extinguisher shall be examined to determine compliance with Section 5 of this specification.

4.6 Examination. The fire extinguisher shall be examined to determine compliance with requirements specified herein. Particular attention should be given, but not limited to, the following:

(a) Overall weight.

(b) Dimensions.

(c) Hose size and lengths.

(d) Arrangement of valves, regulators, gages, plumbing vents.

(e) Type of materials and finishes.

(f) Color.

(g) Legibility and correctness of marking.

(h) Tiedown and lifting means.

(i) Agent tank certifications.

(j) Freedom of valve operation.

(k) Ease of refilling agent tanks and replacing nitrogen cylinders.

(l) Completeness, including caps, funnel and dipstick.

(m) Suitability of electrical heating system.

(n) Free unreeling and rewinding of discharge hose.

(o) Complete drainage of all liquids.

4.7 Tests

4.7.1 Tank pressure test. The AFFF and dry chemical agent tanks shall be pressure tested. Relief valves shall be gagged and openings not related to this test shall be plugged. The tanks shall be pressurized to 150 percent of the 250 psi operating pressure. The pressure shall be held for 15 minutes at both 130°F and -20°F. There shall be no evidence of leakage, permanent distortion or other defects.

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4.7.2 Hose and coupling assembly pressure test. The dry chemical and AFFF discharge hoses and the hose reel and coupling assembly shall be hydrostatically tested to 600 psi pressure for 5 minutes. There shall be no evidence of leakage, permanent distortion or other defects.

4.7.3 High pressure piping and fitting assemblies pressure test. The high pressure piping and fitting assemblies shall be hydraulically tested to 4,000 psi for 1 minute. There shall be no leakage or other defects.

4.7.4 Cycling and rupture test. A sample of both the dry chemical tank and the AFFF tank, selected from the tank manufacturers production run, which is used for fabricating the fire extinguisher, shall be selected for this test. The tank shall be charged from atmospheric pressure to the tank's psi design operating pressure and then below 250 percent of the tank's design operating pressure. Both agent tanks shall be so tested.

4.7.5 Operating tests

4.7.5.1 Pressurization. It shall take no more than 10 seconds for the nitrogen cylinders to develop 80 percent of the design operating pressure in each agent tank after valves have been fully opened. The pressure measurements shall be made with direct reading test gages located near fill openings of the tanks.

4.7.5.2 Pressure relief. The pressure relief valves shall be tested to assure that each does not permit the controlled pressure to build beyond 275 psi, i.e., 110 percent of the designed operating pressure capacity of the tank. Each relief valve shall be tested to assure that it allows full nitrogen flow when relieving. Test gages shall be installed at the fill openings of the tanks.

4.7.5.3 Temperature relief. The temperature relief device installed on the AFFF agent tank shall be tested to assure that it relieves at a temperature greater than 190°F but less than 212°F.

4.7.5.4 Compacted discharge. After filling the dry chemical agent tank, the charge shall be compacted by hauling the complete extinguisher on a truck over 50 miles of gravel road to achieve maximum reduction in volume from the "fresh-fill" condition. After compaction has been achieved, the dry chemical agent tank shall be discharged in a normal manner. The design efficiency shall be such that the powder residue remaining in the dry chemical agent tanks shall not exceed 10 percent by weight of the original charge.

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4.7.5.5 Intermittent discharge. The fully charged and pressurized extinguisher shall be discharged in 5 seconds "ON" and 5 seconds "OFF", alternating bursts until both agents have been exhausted. The design shall be rejected unless the discharge is free at all times and at least 90 percent of the dry chemical capacity has been discharged.

4.7.5.6 Operating characteristics. Both agent tanks shall be charged and pressurized to their design operating pressures and capacities. The tests shall be conducted at ambient temperatures of 65°F to 80°F at less than 5-mph wind velocity. The nozzles shall be pointed about 15° above horizontal at a height of 4 feet above the ground. When tested in accordance with NFPA Pamphlet No. 412, the rate of discharge of the AFFF solution shall be at least 45 gpm. The foam produced shall have a minimum expansion ratio of at least five and a minimum 25 percent drainage time of 1 minute when measured as described in section 422 and Appendix A sections A-220, A-230 and A-240 of NFPA Pamphlet No. 412. The AFFF nozzle ground pattern shall produce a maximum reach of 45 to 50 feet and a nominal width of 12 feet when the nozzle inlet pressure is 100 psi. The rate of discharge of the dry chemical agent shall be at least 4 pounds per second and provide an effective reach of the powder stream of 40 to 50 feet with wind less than 5 mph. A maximum of 90 percent of the dry powder agent shall be discharged at an overall average rate of not less than 4 pounds per second.

4.7.5.7 Plumbing and nozzle leakage. The shutoff valves or nozzles shall effectively prevent the leakage of gas, dry chemical and AFFF during this test. In this test, the fire extinguisher is to be charge with the exact rated capabilities of expellant gas, dry chemical and AFFF active solution and operated intermittently by opening and closing the dry chemical and AFFF discharged valves of nozzles for a period of 5 seconds "open" and allowed to set for 30 minutes "closed". Upon being opened again and being held open until the end of discharge is reached, the total amount of dry chemical discharge shall be not less than 90 percent of the original charge. There shall be no evidence of leaks during this test cycle.

4.7.5.8 AFFF tank heater. THe fully filled and charged fire extinguisher shall be subjected to a -40°F cold chamber test for 8 hours. The thermostatically controlled heater, compatible with 3-phase, 60-Hz, 4-wire (wye) 120-208 Vac source, shall maintain the AFFF solution above 40°F. Operation of the overheat safety device shall be tested to shut off power if solution temperature reaches 80°F.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial, as specified (see 6.2).

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5.1.1 Level A. Preservation and packaging shall be in accordance with MIL-STD-794. Materials used for cleaning, preserving and packaging shall conform to the requirements of MIL-P-116.

5.1.2 Commercial. Fire extinguishers shall be preserved and packaged in accordance with the manufacturer's standard practice, in a manner to insure safe delivery to the destination specified in the contract or order.

5.2 Packing. Packing shall be level A or commercial (see 6.2).

5.2.1 Level A. Fire extinguishers, packaged as specified in 5.1, shall be packed in wood cleated plywood containers conforming to PPP-B-601, overseas type, or nailed wood boxes conforming to PPP-B-621, class 2. The fire extinguishers shall be shipped fully charged with dry chemical, full nitrogen cylinders and a package of AFFF concentrate which will be mixed with water provided at the destination.

5.2.2 Commercial. Fire extinguishers shall be packed in accordance with the manufacturer's standard practice, in a manner to insure safe delivery to the destination specified in the contract or order. The fire extinguishers shall be shipped fully charged with dry chemical, full nitrogen cylinders and a package of AFFF concentrate which will be mixed with water provided at the destination.

5.3 Marking. In addition to any special marking specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fire extinguisher covered by this specification is primarily intended for use in rapidly combating hydrocarbon type fires endangering U.S. Marine Corps amphibious assault fuel system specified in MIL-F-82129.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.

(b) Level of preservation, packaging and packing required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item

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Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement	Applicable DID	Options
4.4	Inspection System Program Plan	DI-R-4803	-
4.3	First Article Inspection Procedure	DI-T-4901	-
4.3	First Article Inspection Report	DI-T-4902	-
3.2.1	Nonstandard Part Approval Requests/ Proposed Additions to an Approved Programs Parts Selection List	DI-E-7028B	-

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 First article. The first article testing shall be in accordance with Section 4.

Preparing activity:

Navy - MC

Project No. 4210-N188

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-E-29406(MC) EXTINGUISHER, FIRE, DRY CHEMICAL AND AQUEOUS FILM, ETC.

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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